Jungwoo Rhee

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KEYWORDS Human-Computer Interaction | Multimodal Interaction | Extended Reality | Design and Prototyping Tools

EDUCATION

KAIST, Daejeon, Korea
- Ph.D. in Industrial Design (Human-Computer Interaction) Sep 2025 – Aug 2029 (expected)
  - Advisor: Prof. Andrea Bianchi
- M.Sc. in Industrial Design (Human-Computer Interaction) Aug 2023 – Aug 2025
  - Thesis: Supporting Designers to Create Environmentally Responsible XR UIs
  - Advisor: Prof. Andrea Bianchi

Kookmin University, Seoul, Korea
- B.F.A. in Spatial Design, Minor in Software Mar 2018 – Aug 2023

Northeastern Illinois University, Chicago, Illinois, U.S.A.
- International Student Exchange Program in Art + Design Aug 2021 – Dec 2021

PUBLICATIONS

JOURNAL PAPERS

[7] Jungwoo Rhee, Guhn Lee, Yumin Jung, Mustafa Doga Dogan, Justin Moon, and Andrea Bianchi, "AdapTest: Supporting Designers in Authoring and Evaluating Environmentally Responsive User Interfaces for Extended Reality," in *Proceedings of the ACM on Human-Computer Interaction (PACM HCI), ISS 2026*, Torino, Italy, Nov 2026. (To appear.)

CONFERENCE FULL PAPERS

[6] Punn Lertjaturaphat*, Jungwoo Rhee*, Jaewon You, and Andrea Bianchi, "Wire Your Way: Hardware-Contextualized Guidance and In-situ Tests for Personalized Circuit Prototyping," in *Proceedings of CHI 2026*, Barcelona, Spain, Apr 2026. **Honorable Mention Award (top 5%)**. (* authors contributed equally.)

[5] Anam Ahmad Khan, Florian Weidner, Jungwoo Rhee, Yasmeen Abdrabou, Andrea Bianchi, Eduardo Velloso, Hans Gellersen, and Joshua Newn, "Gaze and Speech in Multimodal Human-Computer Interaction: A Scoping Review," in *Proceedings of CHI 2026*, Barcelona, Spain, Apr 2026.

CONFERENCE EXTENDED ABSTRACTS AND WORKSHOP PAPERS

[4] Seunghwa Pyo*, Donggun Lee*, Jungwoo Rhee*, Soobin Park, and Youn-kyung Lim, "One Is Not Enough: How People Use Multiple AI Models in Everyday Life," in *Extended Abstracts of CHI 2026*, Barcelona, Spain, Apr 2026. (* authors contributed equally.)

[3] Jungwoo Rhee, "Deepening, Redirecting, Expanding: A Design Space for Generative Interfaces with Qualitative Inquiry," in *CHI 2026 Workshop on Generative UI*, Barcelona, Spain, Apr 2026.

[2] Jungwoo Rhee, "Passive Observation and Asynchronous Feedback to Mitigate Meeting Fatigue," in *CHI 2025 Workshop on Scaling Distributed Collaboration in Mixed Reality*, Yokohama, Japan, Apr 2025.

DOMESTIC CONFERENCE PAPERS

[1] Jungwoo Rhee and Andrea Bianchi, "How Professionals Use Visual Asset Management: An Exploration through Interviews," in *Proceedings of HCI Korea 2024*, Korea, Jan 2024.

MANUSCRIPTS UNDER REVIEW
- Second author: attention-aware code reading support with personalized feedback grounded in expert gaze and code complexity.
- Co-author: contact-adaptive haptic interface with co-located thermal and vibrotactile feedback from a single gel membrane.

PATENTS
- Andrea Bianchi, Jungwoo Rhee, Punn Lertjaturaphat, and Jaewon You, "System and Method for Supporting Circuit Prototyping Based on Circuit Design Information," Korean Patent Application No. 10-2026-0156356.

RESEARCH EXPERIENCES

KAIST, Daejeon, Korea
- Graduate Researcher, make Lab Aug 2023 – Now
  - Supervisor: Prof. Andrea Bianchi.
  - Topics: situated reference in multimodal interaction, context-aware AI support for physical computing, and environmen-

tally adaptive XR interfaces (see publications [7], [6], and [5]).

PROJECTS & GRANTS

- Development and User Evaluation of a Sensory Substitution-Based Interface for Reducing Cognitive Load in Spatial Cognition — Jul 2026 – Now
  - Funded by the Electronics and Telecommunications Research Institute (ETRI).
- Development of a Multi-AI Agent for Autonomous Decision-Making Specialized in the Tofu Coagulation Process — Jun 2026 – Now
  - Funded by the Korea Technology and Information Promotion Agency for SMEs (TIPA).
  - Role: Project lead.
- XR-Based Walking Assistance System: User Testing and Design Workshop — Dec 2025 – Feb 2026
  - Funded by the Electronics and Telecommunications Research Institute (ETRI).
- Soma Design Interaction With Soft Haptics — Nov 2024 – Now
  - Funded by the National Research Foundation of Korea (NRF).
- SpaceTop: Spatial Computing HCI Technology for Everywhere XR Productivity Workstation — Aug 2024 – Now
  - Funded by the Institute of Information & Communications Technology Planning & Evaluation (IITP), ITRC program; 8-year national R&D project.
  - Role: Project lead for the 2nd and 3rd project years.
  - Research on how XR systems can support everyday professional work beyond fixed-place desktop environments.
- Listening Through the Body: Combining Visual, Auditory, and Haptic Interaction While Designing a Concert Space for VR — Jun 2024 – Now
  - Funded by the National Research Foundation of Korea (NRF).
  - Led first-year user experience research and prototype operations for a virtual concert environment; co-organized the NRF-ASPIRE Workshop 2026.
- 3D-Printed Thermoplastic Volumetric Displays With Pixel-Level Control — Jun 2024 – Dec 2024
  - Funded by KAIST (internal research program).
- NH Investment & Securities – KAIST UX Design Research Center — Jan 2024 – Mar 2025
  - Funded by NH Investment & Securities Co., Ltd.
  - Project manager for *Challenge AI for Reflective Investment Decisions*: designing AI agents that help investors reflect before high-stakes financial decisions.
  - Analyzed 40 financial AI services, coordinated a multi-stakeholder design workshop, and evaluated 12 prototypes of challenge-based AI interventions.
- Center for Anthropocene Studies (CAS) — Oct 2023 – Apr 2024
  - Funded by the National Research Foundation of Korea (NRF).
- AutoPalp: Device for Remote Abdominal Examination — Sep 2023 – Feb 2024
  - Funded by the Korea University Medical Center.

HONORS & AWARDS

- Honorable Mention Award, ACM CHI 2026. — Apr 2026
  - Awarded to the top 5% of papers, for *Wire Your Way* (publication [6]).
- Finalist, International Design Excellence Awards (IDEA), Professional Category. — Aug 2025
  - Project: AIRCUBE.
- Ministry of Education Prize. — Aug 2024
  - Promising Student Start-up Team 300+, Value-Up Track.
- Grand Prize, Seoul Open Data Square Mobile App/Web Contest. — Jun 2023
- School of Design Scholarship, Kookmin University. — Mar 2022
  - 50% departmental tuition scholarship for one year.

ACADEMIC SERVICE

- Paper Reviewing.
  - ACM CHI 2027.
  - ACM IMWUT 2026.
  - ACM DIS 2026.
  - PACM HCI (ISS) 2026.
  - ACM VRST 2026.
  - ACM SUI 2026.
- Organization and Volunteering.
  - Co-organizer and Web Chair, NRF-ASPIRE Workshop 2026: Listening Through the Body, Sep 2026.
  - Assistant to the Program Chairs, ACM UIST 2025.
  - Student Volunteer, ACM UIST 2025.
  - Student Volunteer, ACM CHI 2025.

TEACHING

- Teaching Assistant for Software Prototyping (KAIST) — Spring 2025, Spring 2026
- Teaching Assistant for Designing Interactions with AI (KAIST) — Fall 2025
- Teaching Assistant for Korean 2 (KAIST) — Fall 2024
  - School of Digital Humanities and Computational Social Sciences; supported international students learning Korean.
- Teaching Assistant for Graphic Design (KAIST) — Spring 2024

- Teaching Assistant for Interactive Space (KAIST) Fall 2023

INDUSTRY EXPERIENCE

- Product Designer, Fliption Inc. Apr 2025 – Jul 2025
  - Designed the end-to-end user-side experience, interaction flows, and interface concepts for an AI-powered virtual try-on platform.
- Design Tutor, Litas Corp. Mar 2022 – Aug 2023
  - Taught design principles, portfolio development, and visual communication to 200+ high school and undergraduate students.
- Web Design Intern, Cherish Co., Ltd. Dec 2020 – Feb 2021
- Signage Design Research Intern, Bluestone Cooperation May 2019 – Aug 2019
- Content Design Intern, Interstella Inc. Jun 2018 – Aug 2018